

50 Python Pattern Programs

Here are 50 different Python programs to print patterns. Each pattern includes a description and the corresponding Python code. Learn and explore different patterns with CodeWithCurious.

1. Right Angle Triangle Pattern

```
for i in range(1, 6):  
    print('*' * i)
```

```
 *  
* *  
* * *  
* * * *  
* * * * *
```

2. Inverted Right Angle Triangle Pattern

```
for i in range(5, 0, -1):  
    print('*' * i)
```

```
*****  
****  
***  
**  
*
```

3. Pyramid Pattern

```
for i in range(1, 6):  
    print(' ' * (5 - i) + '*' * (2 * i - 1))
```

```
      *  
     * * *  
    * * * * *  
   * * * * * * *  
  * * * * * * * *  
 * * * * * * * * *
```

4. Inverted Pyramid Pattern

```
for i in range(5, 0, -1):  
    print(' ' * (5 - i) + '*' * (2 * i - 1))
```

```
*****  
*****  
*****  
***  
*  

```

5. Diamond Pattern

```
for i in range(1, 6):
```

```

    print(' ' * (5 - i) + '*' * (2 * i - 1))
for i in range(4, 0, -1):
    print(' ' * (5 - i) + '*' * (2 * i - 1))

```

```

      *
     * * *
    * * * * *
   * * * * * * *
  * * * * * * * * *
 * * * * * * * *
* * * * * *
 * * *
  *

```

6. Hollow Square Pattern

```

for i in range(5):
    for j in range(5):
        if i == 0 or i == 4 or j == 0 or j == 4:
            print('*', end='')
        else:
            print(' ', end='')
    print()

```

```

***** * * *
* * * *****

```

7. Full Square Pattern

```
for i in range(5):  
    print('*' * 5)
```

```
*****  *****  
*****  *****  
*****
```

8. Right Angle Triangle (Number Pattern)

```
for i in range(1, 6):  
    print(' '.join(str(x) for x in range(1, i + 1)))
```

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

9. Inverted Right Angle Triangle (Number Pattern)

```
for i in range(5, 0, -1):  
    print(' '.join(str(x) for x in range(1, i + 1)))
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

10. Floyd's Triangle

```
num = 1
for i in range(1, 6):
    for j in range(1, i + 1):
        print(num, end=' ')
        num += 1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

11. Hollow Right Angle Triangle

```
for i in range(1, 6):
    for j in range(1, i + 1):
        if j == 1 or j == i or i == 5:
            print('*', end='')
        else:
            print(' ', end='')
    print()
```

```

*
*
*
*  *
*****
*
```

12. Hollow Pyramid Pattern

```

for i in range(1, 6):
    for j in range(5 - i):
        print(' ', end='')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2 or i == 5:
            print('*', end='')
        else:
            print(' ', end='')
    print()
```

```

    *
  * *
 *   *
*     *
*****
```

13. Hollow Diamond Pattern

```

n = 5
for i in range(1, n + 1):
```

```

print(' ' * (n - i) + '*' + ' ' * (2 * i - 3) + ('*' if i > 1 else ''))
for i in range(n - 1, 0, -1):
    print(' ' * (n - i) + '*' + ' ' * (2 * i - 3) + ('*' if i > 1 else ''))

```

```

      *
     * *
    *  *
   *    *
  *      *
 *****
  *      *
   *    *
    *  *
     * *
      *

```

14. Hollow Diamond (Number Pattern)

```

n = 5
for i in range(1, n + 1):
    for j in range(1, n - i + 1):
        print(' ', end='')
    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1:
            print(i, end='')
        else:
            print(' ', end='')
    print()

```

```

      1
     2 2
    3  3
   4    4
  5      5
 4      4
 3  3
 2 2

```

15. Butterfly Pattern

```
for i in range(1, 6):
    for j in range(1, i + 1):
        print('*', end='')
    for j in range(1, 2 * (5 - i)):
        print(' ', end='')
    for j in range(1, i + 1):
        print('*', end='')
    print()
```

```

* *  ** **  ***
***      ****  ****
*****      ****
****  ***  ***  **
**  * *
```

16. Hollow Number Pyramid

```
for i in range(1, 6):
    for j in range(1, 6 - i):
        print(' ', end='')
    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1:
            print(i, end='')
        else:
            print(' ', end='')
    print()
```

```
else:
    print(' ', end='')
print()
```

```
    1
  2 2
 3  3
4   4
5   5
```

17. Full Star Pyramid

```
for i in range(1, 6):
    print(' ' * (5 - i) + '*' * (2 * i - 1))
```

```
    *
  * * *
* * * * *
* * * * * * *
* * * * * * * *
```

18. Inverted Full Star Pyramid

```
for i in range(5, 0, -1):
    print(' ' * (5 - i) + '*' * (2 * i - 1))
```

```
*****
*****
```

```
*****
***
*
```

19. Left Aligned Pyramid Pattern

```
for i in range(1, 6):
    print('*' * i)
```

```
*
* *
* * *
* * * *
* * * * *
```

20. Right Aligned Pyramid Pattern

```
for i in range(1, 6):
    print(' ' * (5 - i) + '*' * i)
```

```

*      * *
* * *
* * * *
* * * * *
```

21. Pascal's Triangle

```
n = 5
for i in range(n):
    for j in range(n - i - 1):
        print(' ', end='')
    for j in range(i + 1):
        print(str(math.comb(i, j)) + ' ', end='')
    print()
```

```
      1
     1 1
    1 2 1
   1 3 3 1
  1 4 6 4 1
```

22. Zig-Zag Pattern

```
for i in range(1, 6):
    if i % 2 == 0:
        print(' ' * (i - 1) + '*')
    else:
        print('*' * i)
```

```
  *
 *
*
 *
*
```

23. Hourglass Pattern

```
for i in range(5, 0, -1):
    print(' ' * (5 - i) + '*' * (2 * i - 1))
for i in range(2, 6):
    print(' ' * (5 - i) + '*' * (2 * i - 1))
```

```
*****
*****
*****
***
*
***
*****
*****
*****
```

24. Diamond Shape with Numbers

```
for i in range(1, 6):
    print(' ' * (5 - i) + str(i) * (2 * i - 1))
for i in range(4, 0, -1):
    print(' ' * (5 - i) + str(i) * (2 * i - 1))
```

```
  1
 2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
5 5 5 5 5 5 5 5
4 4 4 4 4 4 4
3 3 3 3 3
 2 2 2
  1
```

25. Hollow Rhombus Pattern

```
for i in range(1, 6):
    for j in range(1, 6):
        if i == 1 or i == 5 or j == 1 or j == 5:
            print('*', end='')
        else:
            print(' ', end='')
    print()
```

```
***** * * *
* * * *****
```

26. Numeric Pyramid

```
for i in range(1, 6):
    print(' ' * (5 - i) + ' '.join(str(x) for x in range(1, i + 1)))
```

```
      1
     1 2
    1 2 3
   1 2 3 4
  1 2 3 4 5
```

27. Hollow Diamond with Numbers

```
for i in range(1, 6):
    for j in range(1, 6 - i):
        print(' ', end='')
    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1:
            print(i, end='')
        else:
            print(' ', end='')
    print()
for i in range(4, 0, -1):
    for j in range(1, 6 - i):
        print(' ', end='')
    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1:
            print(i, end='')
        else:
            print(' ', end='')
    print()
```

```
      1
     2 2
    3   3
   4     4
  5       5
 4        4
 3       3
 2 2
 1
```

28. Reverse Pyramid with Numbers

```
for i in range(5, 0, -1):
    print(' ' * (5 - i) + ' '.join(str(x) for x in range(1, i + 1)))
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

29. Diamond Star Pattern

```
for i in range(1, 6):
    print(' ' * (5 - i) + '*' * (2 * i - 1))
for i in range(4, 0, -1):
    print(' ' * (5 - i) + '*' * (2 * i - 1))
```

```
      *
    * * *
  * * * * *
* * * * * * *
* * * * * * * *
* * * * * * * *
* * * * *
  * * *
    *
```

30. Full Number Pyramid

```
for i in range(1, 6):
    print(' ' * (5 - i) + ''.join(str(x) for x in range(1, i + 1)))
```

```
    1      1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

31. Checkerboard Pattern

```
for i in range(5):
    for j in range(5):
        if (i + j) % 2 == 0:
            print('*', end='')
        else:
            print(' ', end='')
    print()
```

```
* * *
 * *
* * *
 * *
* * *
```

32. Hollow Circle Pattern

```
import math
for i in range(1, 6):
    for j in range(1, 6):
        if math.dist([i, j], [3, 3]) <= 2:
            print('*', end='')
        else:
```

```
        print(' ', end='')
print()
```

```
        *****
      *       *
    *         *
  *           *
*             *
*****
```

33. Triangle of Numbers

```
for i in range(1, 6):
    for j in range(1, i + 1):
        print(j, end='')
    print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

34. Rhombus of Numbers

```
for i in range(1, 6):
    print(' ' * (5 - i) + ''.join(str(x) for x in range(1, i + 1)))
for i in range(4, 0, -1):
    print(' ' * (5 - i) + ''.join(str(x) for x in range(1, i + 1)))
```

```

    1      1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4
1 2 3
1 2      1

```

35. Number Pattern (Decreasing)

```

for i in range(5, 0, -1):
    print(' '.join(str(x) for x in range(1, i + 1)))

```

```

1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

```

36. Hollow Inverted Pyramid

```

for i in range(5, 0, -1):
    for j in range(1, 2 * i - 1):
        if j == 1 or j == 2 * i - 2 or i == 5:
            print('*', end='')
        else:
            print(' ', end='')
    print()

```

```

*****
 *      *
  *    *
   *  *
    *

```

37. Cross Pattern

```

for i in range(5):
    for j in range(5):
        if i == j or i + j == 4:
            print('*', end='')
        else:
            print(' ', end='')
    print()

```

```

 *    *
  *  *
   *
  *  *
 *    *

```

38. Number Inverted Pyramid

```

for i in range(5, 0, -1):
    print(' '.join(str(x) for x in range(1, i + 1)))

```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

39. Right Angle Star Triangle

```
for i in range(1, 6):
    print(' ' * (5 - i) + '*' * i)
```

```
    *      * *
* * *
* * * *
* * * * *
```

40. Left Angle Number Triangle

```
for i in range(1, 6):
    print(' '.join(str(x) for x in range(1, i + 1)))
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

41. Star Hollow Triangle

```
for i in range(1, 6):
    for j in range(1, 6):
        if j == 1 or j == i or i == 5:
            print('*', end='')
        else:
            print(' ', end='')
    print()
```

```
  *
 * *
*   *
 * *
*****
```

42. Upside Down Right Triangle

```
for i in range(5, 0, -1):
    print(' ' * (5 - i) + '*' * i)
```

```
  *****
*** ** *
```

43. Parallelogram Star Pattern

```
for i in range(5):  
    print(' ' * i + '*' * 5)
```

```
*****  
 *****  
  *****  
   *****  
    *****
```

44. Inverted Parallelogram Pattern

```
for i in range(5):  
    print(' ' * (5 - i) + '*' * 5)
```

```
*****  
 *****  
  *****  
   *****  
    *****
```

45. Reverse Hollow Diamond

```
for i in range(5, 0, -1):  
    print(' ' * (5 - i) + '*' + ' ' * (2 * i - 3) + ('*' if i > 1 else ''))  
for i in range(4, 0, -1):  
    print(' ' * (5 - i) + '*' + ' ' * (2 * i - 3) + ('*' if i > 1 else ''))
```

```

*****
 *    *
  *  *
   *
  *
 *  *
 *    *
*****

```

46. Inverted Hollow Pyramid with Stars

```

for i in range(5, 0, -1):
    for j in range(1, 2 * i - 1):
        if j == 1 or j == 2 * i - 2 or i == 5:
            print('*', end='')
        else:
            print(' ', end='')
    print()

```

```

*****
 *    *
  *  *
   *
  *
 *    *
*****

```

47. Full Hollow Diamond

```

for i in range(1, 6):

```

```

for j in range(1, 6 - i):
    print(' ', end='')
for j in range(1, 2 * i):
    if j == 1 or j == 2 * i - 1:
        print('*', end='')
    else:
        print(' ', end='')
print()
for i in range(4, 0, -1):
    for j in range(1, 6 - i):
        print(' ', end='')
    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1:
            print('*', end='')
        else:
            print(' ', end='')
    print()

```

```

      1
     2 2
    3   3
   4     4
  5       5
 4        4
 3       3
 2 2
 1

```

48. Rectangle of Numbers

```

for i in range(5):
    for j in range(1, 6):
        print(j, end=' ')
    print()

```

```

1 2 3 4 5

```

```
1 2 3 4 5 1 2 3 4
5 1 2 3 4 5 1 2 3
4 5
```

49. Reverse Star Pattern

```
for i in range(5, 0, -1):
    print('*' * i)
```

```
*****
****
***
**
*
```

50. Pyramid of Numbers

```
for i in range(1, 6):
    print(' ' * (5 - i) + ''.join(str(x) for x in range(1, i + 1)))
```

```
1      1 2
1 2 3
1 2 3 4
1 2 3 4 5
```